Cletus: Section 1, Day 4

- Date: [YYYY-MM-DD]
- Course: Week 1 Day 4, transformers, parameters, tokens, the illusion of memory, context windows and API costs
- Lab notebook: week1/day4.ipynb
- Time spent: [e.g. 2h]

Key concepts

- Transformer: the architecture behind GPT and nearly every modern LLM. It comes from Google's 2017 paper "Attention Is All You Need". Older models (RNNs and LSTMs) read text one word at a time. Attention lets the model look at all the words at once and work out which ones matter, so it can be trained in parallel on huge amounts of data.
- Emergent abilities: once models get big enough, they can do things nobody explicitly trained them to do.
- Parameters (weights): the numbers a model learns in training, basically its size. GPT-1 had about 117 million, GPT-3 had 175 billion, Llama 3.1 goes up to 405 billion and DeepSeek V3/R1 has 671 billion. The top closed models are believed to be in the trillions.
- Tokens: the chunks of text a model reads and writes. They're not letters, and not always whole words. In English a token is roughly 4 characters, or about three quarters of a word, so 1,000 tokens is around 750 words. Common words are a single token, while rare words and names get split up. The space before a word is often part of the token (" pie").
- Tokenizer: converts text to token IDs and back. Every model family has its own, and `tiktoken` is OpenAI's.
- Stateless calls (the illusion of memory): each API call starts from zero. The model only seems to remember because we send the whole conversation again every time, including its own earlier replies with the `assistant` role. ChatGPT does exactly this.
- Context window: the maximum number of tokens the model can handle in one call. That covers the system prompt, the whole conversation and the reply. Very long chats eventually run into it.
- API costs: you pay per token, usually quoted per million. Input and output have separate prices, and output is more expensive. Since the conversation is resent each turn, you pay for it again each turn.

Code / examples

Looking at the tokens:

```
import tiktoken

encoding = tiktoken.encoding_for_model("gpt-4.1-mini")
tokens = encoding.encode("Hi my name is Ed and I like banoffee pie")
for token_id in tokens:
    print(token_id, "=", repr(encoding.decode([token_id])))
print(len(tokens), "tokens")
```

Memory is just resending the history:

```
messages = [
    {"role": "system", "content": "You are a helpful assistant"},
    {"role": "user", "content": "Hi! I'm Ed!"},
```

```
    {"role": "assistant", "content": "Hi Ed! How can I assist you today?"},
    {"role": "user", "content": "What's my name?"},
]
# If we only send the last user message, the model has no way of knowing the name.
# With the full history it answers "Ed".
```

Exercise / challenge

1. Tokenize your full name, a sentence in another language and a line of code. Which one uses the most tokens per word?
2. Build a small chatbot that keeps its memory by adding every turn to `messages`:

```
from openai import OpenAI
openai = OpenAI()

messages = [{"role": "system", "content": "You are a helpful assistant"}]
while True:
    user = input("You: ")
    if user == "quit":
        break
    messages.append({"role": "user", "content": user})
    reply = openai.chat.completions.create(model="gpt-4.1-nano",
messages=messages).choices[0].message.content
    messages.append({"role": "assistant", "content": reply})
    print("AI:", reply)
```

3. After a few turns, count the tokens in all of `messages` and see how the cost grows with every turn.

Questions & blockers

Things to watch out for: - Models are bad at counting letters because they see tokens, not letters. - Long chats get more expensive with every turn and can eventually go past the context window.

To discuss with my partner

- How could we keep a long chat cheap? For example by summarizing older turns, or only keeping the last few messages.
- Try explaining attention to each other in one sentence.

Resources

- OpenAI tokenizer playground: https://platform.openai.com/tokenizer
- The "Attention Is All You Need" paper: https://arxiv.org/abs/1706.03762
- Model prices: https://openai.com/api/pricing